Peter Karlos

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Education

University of Illinois Urbana-Champaign — Champaign, IL
Bachelor of Science in Computer Engineering (GPA: 3.68) — *Expected Graduation: Dec. 2027*

Experience

Course Assistant — ECE 391: Computer Systems Engineering — Aug. 2026 - Present
University of Illinois Urbana-Champaign — *Champaign, IL*
- Supported students in operating systems development, debugging RISC-V assembly and C during office hours.

Software Engineering Intern — May 2026 - Aug. 2026
CACI International Inc. — *Lisle, IL*
- Researched and developed security measures against 15 major system misconfigurations and Linux kernel CVEs across a 3-node Kubernetes cluster using eBPF programs, kernel modules, and bash scripts.
- Developed an automatic package management system for deploying test rootkits through Kali Linux in C with 7 unique, modular exploits using agile and scrum development methodologies.
- Assisted in development of a multi-daemon distributed rootkit detection program in Python, specialized in aggregating and analyzing telemetry to detect containerized rootkits in Kubernetes systems.

Student Engineer — Aug. 2025 - Present
Illinois Space Society – Spaceshot Avionics — *Champaign, IL*
- Designed and routed a custom power management PCB on KiCad using a Bluetooth low energy microcontroller to provide wireless control and real-time system state monitoring across a multi-stage rocket's subsystems.
- Programmed embedded firmware on an STM32 microcontroller in C to communicate with internal components through UART and I2C protocols and facilitate Bluetooth connection to external devices.
- Selected components, soldered PCBs, and verified power delivery and board functionality via electrical testing.

IT Systems Engineering Intern — Jun. 2025 - Aug. 2025
7 Layer Solutions Inc. — *Schaumburg, IL*
- Provided Tier 1 & 2 IT support for 200 users, resolving issues across Microsoft 365, AD, and network systems.
- Assisted migration of 10+ AD servers to Microsoft 365, improving endpoint security through Azure AD and MFA.

Projects

TriangleOS – RISC-V 64-bit Kernel | *C, RISC-V*
- Built a RISC-V OS in C, featuring system calls, drivers, threading, a custom filesystem, and exception handling.
- Engineered a process scheduler and Sv39 virtual memory paging to support concurrent userspace execution.

Pipelined RV32I Processor | *SystemVerilog, RISC-V, RTL, Verdi*
- Designed a 5-stage RISC-V processor in SystemVerilog with branch prediction, hazard stalling, and memory.
- Verified instruction execution using RVFI reference checks and debugged datapath and pipeline timing in Verdi.

Parallel CNN Forward Pass Layer | *C++, CUDA, Nsight*
- Parallelized a CNN forward pass in CUDA, accelerating computations by 200% against un-optimized GPU code.
- Optimized compute throughput and memory bandwidth by implementing shared memory tiling and tensor cores.

FPGA HDMI Text Controller | *SystemVerilog, C, RTL, Vivado, Vitis*
- Developed an AXI4-Lite graphics peripheral with memory-mapped VRAM for 2400-character HDMI text display.
- Integrated MicroBlaze software control, font rendering, and colors; verified AXI transactions in simulation.

Relevant Coursework

ECE 411: Computer organization & design covering processor architecture, memory systems, and digital logic.
ECE 391: Low-level systems programming covering operating systems concepts and OS development in C and RISC-V .
ECE 408: Parallel programming with GPU computing, CUDA, and performance optimization on many-core systems.

Technical Skills

Programming Languages: C/C++, Python, CUDA, Verilog/SystemVerilog, Java, RISC-V, JavaScript, HTML/CSS
Software & Libraries: Vivado, Vitis, Verdi, PyTorch, Docker, React, WebGL 2, Kali Linux, Kubernetes, eBPF
Hardware & Tools: RTL Design, Arduino, Raspberry Pi, STM32, Spartan FPGA, KiCad, LTspice, Nsight, Fusion 360